1)

$y = \sinh x$

Swap x and y:

$x = \sinh y$

$x = \frac{e^y - e^{-y}}{2}$

$2x = e^y - e^{-y}$

$2xe^y = e^{2y} - 1$

$0 = e^{2y} - 2xe^y - 1$

$0 = (e^y - x)^2 - x^2 - 1$

$\Rightarrow (e^y - x)^2 = x^2 + 1$

$\Rightarrow e^y - x = \pm\sqrt{x^2 + 1}$

$\Rightarrow e^y = x \pm \sqrt{x^2 + 1}$

Now $e^y > 0$ and $x - \sqrt{x^2 + 1} < 0$, so

$e^y = x + \sqrt{x^2 + 1}$

$\Rightarrow y = \ln\left(x + \sqrt{x^2 + 1}\right)$

$\therefore \operatorname{arsinh} x = \ln\left(x + \sqrt{x^2 + 1}\right)$

2)

$y = \cosh x$ is many-to-one, so we take $x > 0$ to restrict the domain to make it one-to-one.

Swap x and y:

$x = \cosh y$ where $y > 0$

$x = \frac{e^y + e^{-y}}{2}$

$2x = e^y + e^{-y}$

$2xe^y = e^{2y} + 1$

$0 = e^{2y} - 2xe^y + 1$

$0 = (e^y - x)^2 - x^2 + 1$

$\Rightarrow (e^y - x)^2 = x^2 - 1$

$\Rightarrow e^y - x = \pm\sqrt{x^2 - 1}$

$\Rightarrow e^y = x \pm \sqrt{x^2 - 1}$

$\Rightarrow y = \ln\left(x \pm \sqrt{x^2 - 1}\right)$

Now, $\left(x + \sqrt{x^2 - 1}\right)\left(x - \sqrt{x^2 - 1}\right) \equiv x^2 - (x^2 - 1) \equiv 1$

So $x - \sqrt{x^2 - 1} \equiv \frac{1}{x + \sqrt{x^2 - 1}} \equiv \left(x + \sqrt{x^2 - 1}\right)^{-1}$

Then $\ln\left(x - \sqrt{x^2 - 1}\right) \equiv \ln\left(x + \sqrt{x^2 - 1}\right)^{-1} \equiv -\ln\left(x + \sqrt{x^2 - 1}\right)$

$\Rightarrow y = \pm \ln\left(x + \sqrt{x^2 - 1}\right)$

As $y > 0$, we must have that $y = \ln\left(x + \sqrt{x^2 - 1}\right)$

$\therefore \operatorname{arcosh} x = \ln\left(x + \sqrt{x^2 - 1}\right)$ where $x \geq 1$

3)

$y = \tanh x$

Swap x and y:

$x = \tanh y$

$x = \frac{e^{2y}-1}{e^{2y}+1}$

$x(e^{2y}+1) = e^{2y} - 1$

$xe^{2y} + x = e^{2y} - 1$

$x + 1 = e^{2y} - xe^{2y}$

$1 + x = (1 - x)e^{2y}$

$\Rightarrow e^{2y} = \frac{1+x}{1-x}$

$\Rightarrow 2y = \ln\left(\frac{1+x}{1-x}\right)$

$\Rightarrow y = \frac{1}{2}\ln\left(\frac{1+x}{1-x}\right)$

$\therefore \operatorname{artanh} x = \frac{1}{2}\ln\left(\frac{1+x}{1-x}\right)$ where $-1 < x < 1$

4) a)

$\operatorname{arsinh} 1 = \ln\left(1 + \sqrt{1^2 + 1}\right) = \ln\left(1 + \sqrt{2}\right)$

4) b)

$\cosh^{-1} 2 = \ln\left(2 + \sqrt{2^2 - 1}\right) = \ln\left(2 + \sqrt{3}\right)$

4) c)

$\operatorname{artanh} \frac{1}{3} = \frac{1}{2}\ln\left(\frac{1+\frac{1}{3}}{1-\frac{1}{3}}\right) = \frac{1}{2}\ln 2$

4) d)

$\sinh^{-1}(-4) = \ln\left(-4 + \sqrt{(-4)^2 + 1}\right) = \ln\left(-4 + \sqrt{17}\right)$

4) e)

$\operatorname{arcosh} 9 = \ln\left(9 + \sqrt{9^2 - 1}\right) = \ln\left(9 + \sqrt{80}\right) = \ln\left(9 + 4\sqrt{5}\right)$

4) f)

$$\tanh^{-1}\left(-\frac{1}{2}\right) = \frac{1}{2}\ln\left(\frac{1+\left(-\frac{1}{2}\right)}{1-\left(-\frac{1}{2}\right)}\right) = \frac{1}{2}\ln\frac{1}{3} = -\frac{1}{2}\ln 3$$

5) a)

$$\sinh x = 0$$
$$\Rightarrow x = \operatorname{arsinh} 0$$
$$= \ln\left(0 + \sqrt{0^2 + 1}\right)$$
$$= \ln 1$$
$$= 0$$

5) b)

$$\sinh x = 3$$
$$\Rightarrow x = \operatorname{arsinh} 3$$
$$= \ln\left(3 + \sqrt{3^2 + 1}\right)$$
$$= \ln\left(3 + \sqrt{10}\right)$$

5) c)

$$\sinh x = -6$$
$$\Rightarrow x = \operatorname{arsinh}(-6)$$
$$= \ln\left(-6 + \sqrt{(-6)^2 + 1}\right)$$
$$= \ln\left(-6 + \sqrt{37}\right)$$

6) a)

$$\cosh x = 1$$
$$\Rightarrow x = \pm \operatorname{arcosh} 1$$
$$= \pm \ln\left(1 + \sqrt{1^2 - 1}\right)$$
$$= \pm \ln 1$$
$$= 0$$

6) b)

$\cosh x = 6$

$\Rightarrow x = \pm \operatorname{arcosh} 6$

$= \pm \ln\left(6 + \sqrt{6^2 - 1}\right)$

$= \pm \ln\left(6 + \sqrt{35}\right)$

6) c)

$\cosh x = \frac{3}{2}$

$\Rightarrow x = \pm \operatorname{arcosh} \frac{3}{2}$

$= \pm \ln\left(\frac{3}{2} + \sqrt{\left(\frac{3}{2}\right)^2 - 1}\right)$

$= \pm \ln\left(\frac{3}{2} + \sqrt{\frac{5}{4}}\right)$

$= \pm \ln\left(\frac{3+\sqrt{5}}{2}\right)$

7) a)

$\tanh x = 0$

$\Rightarrow x = \operatorname{artanh} 0$

$= \frac{1}{2} \ln\left(\frac{1+0}{1-0}\right)$

$= \frac{1}{2} \ln 1$

$= 0$

7) b)

$\tanh x = \frac{1}{4}$

$\Rightarrow x = \operatorname{artanh} \frac{1}{4}$

$= \frac{1}{2} \ln\left(\frac{1+\frac{1}{4}}{1-\frac{1}{4}}\right)$

$= \frac{1}{2} \ln \frac{5}{3}$

7) c)

$\tanh x = -\frac{5}{6}$

$\Rightarrow x = \operatorname{artanh}\left(-\frac{5}{6}\right)$

$= \frac{1}{2}\ln\left(\frac{1+\left(-\frac{5}{6}\right)}{1-\left(-\frac{5}{6}\right)}\right)$

$= \frac{1}{2}\ln\frac{1}{11}$

$= -\frac{1}{2}\ln 11$

8) a)

$\sinh 2x = 5$

$\Rightarrow 2x = \operatorname{arsinh} 5$

$\Rightarrow 2x = \ln\left(5 + \sqrt{5^2 + 1}\right)$

$\Rightarrow x = \frac{1}{2}\ln\left(5 + \sqrt{26}\right)$

8) b)

$\cosh 3x = 8$

$\Rightarrow 3x = \pm \operatorname{arcosh} 8$

$\Rightarrow 3x = \pm \ln\left(8 + \sqrt{8^2 - 1}\right)$

$\Rightarrow x = \pm\frac{1}{3}\ln\left(8 + \sqrt{63}\right)$

$\Rightarrow x = \pm\frac{1}{3}\ln\left(8 + 3\sqrt{7}\right)$

8) c)

$5\tanh(3x - 1) = 2$

$\Rightarrow \tanh(3x - 1) = \frac{2}{5}$

$\Rightarrow 3x - 1 = \operatorname{artanh}\frac{2}{5}$

$\Rightarrow 3x - 1 = \frac{1}{2}\ln\left(\frac{1+\frac{2}{5}}{1-\frac{2}{5}}\right)$

$\Rightarrow 3x - 1 = \frac{1}{2}\ln\frac{7}{3}$

$\Rightarrow x = \frac{1}{3}\left(\frac{1}{2}\ln\frac{7}{3} + 1\right)$

$\Rightarrow x = \frac{1}{6}\ln\frac{7}{3} + \frac{1}{3}$

8) d)

$3\sinh\left(2-\frac{1}{3}x\right)=1$

$\Rightarrow \sinh\left(2-\frac{1}{3}x\right)=\frac{1}{3}$

$\Rightarrow 2-\frac{1}{3}x=\operatorname{arsinh}\frac{1}{3}$

$\Rightarrow 2-\frac{1}{3}x=\ln\left(\frac{1}{3}+\sqrt{\left(\frac{1}{3}\right)^2+1}\right)$

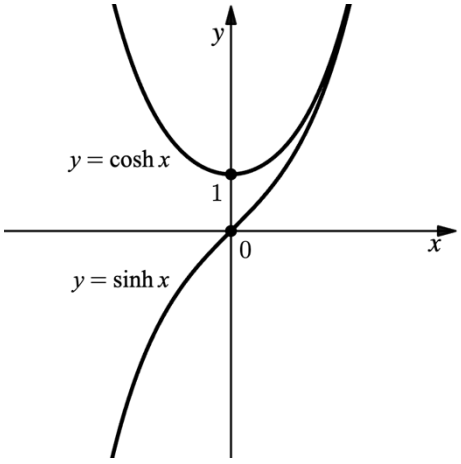
$\Rightarrow 2-\frac{1}{3}x=\ln\left(\frac{1}{3}+\sqrt{\frac{10}{9}}\right)$

$\Rightarrow 2-\frac{1}{3}x=\ln\left(\frac{1+\sqrt{10}}{3}\right)$

$\Rightarrow x=3\left(2-\ln\left(\frac{1+\sqrt{10}}{3}\right)\right)$

$\Rightarrow x=6-3\ln\left(\frac{1+\sqrt{10}}{3}\right)$

9) a)



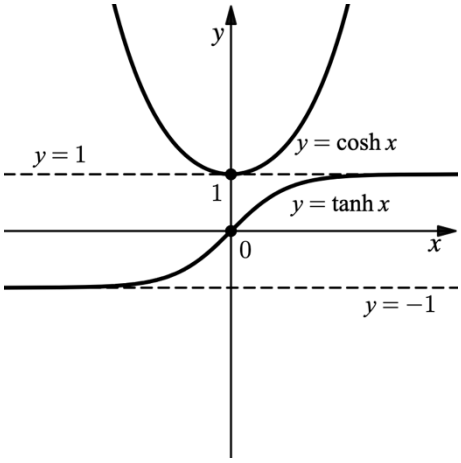
9) b)

$y=\sinh x$ and $y=\cosh x$ intersect if there are solutions to $\sinh x=\cosh x$

$\Rightarrow \frac{\sinh x}{\cosh x}=1 \Rightarrow \tanh x=1 \Rightarrow x=\operatorname{artanh}1=\frac{1}{2}\ln\left(\frac{1+1}{1-1}\right)$ which is undefined

So $y=\sinh x$ and $y=\cosh x$ do not intersect

10) a)



10) b)

$y = \cosh x$ and $y = \tanh x$ intersect if there are solutions to $\cosh x = \tanh x$

$\Rightarrow \cosh x = \frac{\sinh x}{\cosh x}$

$\Rightarrow \cosh^2 x = \sinh x$

$\Rightarrow 1 + \sinh^2 x = \sinh x$

$\Rightarrow \sinh^2 x - \sinh x + 1 = 0$

$\Rightarrow \left(\sinh x - \frac{1}{2}\right)^2 - \left(\frac{1}{2}\right)^2 + 1 = 0$

$\Rightarrow \left(\sinh x - \frac{1}{2}\right)^2 = -\frac{3}{4}$ which has no real solutions.

So $y = \cosh x$ and $y = \tanh x$ do not intersect

11) a)

$\sinh^2 x + \sinh x - 12 = 0$

$\Rightarrow (\sinh x + 4)(\sinh x - 3) = 0$

$\Rightarrow \sinh x = -4$ or $\sinh x = 3$

If $\sinh x = -4$

$\Rightarrow x = \operatorname{arsinh}(-4)$

$= \ln\left(-4 + \sqrt{(-4)^2 + 1}\right)$

$= \ln\left(-4 + \sqrt{17}\right)$

If $\sinh x = 3$

$\Rightarrow x = \operatorname{arsinh} 3$

$= \ln\left(3 + \sqrt{3^2 + 1}\right)$

$= \ln\left(3 + \sqrt{10}\right)$

$\therefore x = \ln\left(-4 + \sqrt{17}\right)$ or $x = \ln\left(3 + \sqrt{10}\right)$

11) b)

$2\cosh^2 x - \cosh x - 21 = 0$

$\Rightarrow (2\cosh x - 7)(\cosh x + 3) = 0$

$\Rightarrow \cosh x = \frac{7}{2}$ or $\cosh x = -3$

$\cosh x = -3$ has no real solutions

$\cosh x = \frac{7}{2}$

$\Rightarrow x = \pm \operatorname{arcosh} \frac{7}{2}$

$= \pm \ln\left(\frac{7}{2} + \sqrt{\left(\frac{7}{2}\right)^2 - 1}\right)$

$= \pm \ln\left(\frac{7}{2} + \sqrt{\frac{45}{4}}\right)$

$= \pm \ln\left(\frac{7+3\sqrt{5}}{2}\right)$

$\therefore x = \pm \ln\left(\frac{7+3\sqrt{5}}{2}\right)$

11) c)

$12\tanh^2 x + \tanh x - 1 = 0$

$\Rightarrow (4\tanh x - 1)(3\tanh x + 1) = 0$

$\Rightarrow \tanh x = \frac{1}{4}$ or $\tanh x = -\frac{1}{3}$

If $\tanh x = \frac{1}{4}$

$\Rightarrow x = \operatorname{artanh}\frac{1}{4}$

$= \frac{1}{2}\ln\left(\frac{1+\frac{1}{4}}{1-\frac{1}{4}}\right)$

$= \frac{1}{2}\ln\frac{5}{3}$

If $\tanh x = -\frac{1}{3}$

$\Rightarrow x = \operatorname{artanh}\left(-\frac{1}{3}\right)$

$= \frac{1}{2}\ln\left(\frac{1+\left(-\frac{1}{3}\right)}{1-\left(-\frac{1}{3}\right)}\right)$

$= \frac{1}{2}\ln\frac{1}{2}$

$= -\frac{1}{2}\ln 2$

$\therefore x = \frac{1}{2}\ln\frac{5}{3}$ or $x = -\frac{1}{2}\ln 2$

12)

$3\cosh^2 x + 10\cosh x - 25 = 0$

$\Rightarrow (3\cosh x - 5)(\cosh x + 5) = 0$

$\Rightarrow \cosh x = \frac{5}{3}$ or $\cosh x = -5$

$\cosh x = -5$ has no real solutions

$\cosh x = \frac{5}{3}$

$\Rightarrow x = \pm\operatorname{arcosh}\frac{5}{3}$

$= \pm\ln\left(\frac{5}{3} + \sqrt{\left(\frac{5}{3}\right)^2 - 1}\right)$

$= \pm\ln\left(\frac{5}{3} + \frac{4}{3}\right)$

$= \pm\ln 3$

So $x = \ln 3$ or $x = -\ln 3 = \ln\frac{1}{3}$

13)

$3\cosh^2 x - 17\cosh x - 56 = 0$

$\Rightarrow (3\cosh x + 7)(\cosh x - 8) = 0$

$\Rightarrow \cosh x = 8$ or $\cosh x = -\frac{7}{3}$

$\cosh x = -\frac{7}{3}$ has no real solutions

$\cosh x = 8$

$\Rightarrow x = \pm \operatorname{arcosh} 8$

$= \pm \ln\left(8 + \sqrt{8^2 - 1}\right)$

$= \pm \ln\left(8 + 3\sqrt{7}\right)$

So $x = \ln\left(8 + 3\sqrt{7}\right)$ or $x = -\ln\left(8 + 3\sqrt{7}\right) = \ln\left(\frac{1}{8+3\sqrt{7}}\right) = \ln\left(8 - 3\sqrt{7}\right)$

$\therefore x = \ln\left(8 + 3\sqrt{7}\right)$ or $x = \ln\left(8 - 3\sqrt{7}\right)$

14) a)

$\operatorname{arsinh} x = \ln 2$

$\Rightarrow \ln\left(x + \sqrt{x^2 + 1}\right) = \ln 2$

$\Rightarrow x + \sqrt{x^2 + 1} = 2$

$\Rightarrow \sqrt{x^2 + 1} = 2 - x$

$\Rightarrow x^2 + 1 = (2 - x)^2$

$\Rightarrow x^2 + 1 = x^2 - 4x + 4$

$\Rightarrow 1 = -4x + 4$

$\Rightarrow 4x = 3$

$\Rightarrow x = \frac{3}{4}$

Alternative Method:

$\operatorname{arsinh} x = \ln 2$

$\Rightarrow x = \sinh(\ln 2)$

$= \frac{e^{\ln 2} - e^{-\ln 2}}{2}$

$= \frac{e^{\ln 2} - e^{\ln \frac{1}{2}}}{2}$

$= \frac{2 - \frac{1}{2}}{2} = \frac{3}{4}$

14) b)

$\operatorname{arcosh} x = \ln 3$

$\Rightarrow \ln\left(x + \sqrt{x^2 - 1}\right) = \ln 3$

$\Rightarrow x + \sqrt{x^2 - 1} = 3$

$\Rightarrow \sqrt{x^2 - 1} = 3 - x$

$\Rightarrow x^2 - 1 = (3 - x)^2$

$\Rightarrow x^2 - 1 = x^2 - 6x + 9$

$\Rightarrow -1 = -6x + 9$

$\Rightarrow 6x = 10$

$\Rightarrow x = \frac{5}{3}$

Alternative Method:

$\operatorname{arcosh} x = \ln 3$

$\Rightarrow x = \cosh(\ln 3)$

$= \frac{e^{\ln 3} + e^{-\ln 3}}{2}$

$= \frac{e^{\ln 3} + e^{\ln \frac{1}{3}}}{2}$

$= \frac{3 + \frac{1}{3}}{2} = \frac{5}{3}$

14) c)

$\operatorname{artanh} x = \ln 4$

$\Rightarrow \frac{1}{2}\ln\left(\frac{1+x}{1-x}\right) = \ln 4$

$\Rightarrow \ln\left(\frac{1+x}{1-x}\right) = 2\ln 4$

$\Rightarrow \ln\left(\frac{1+x}{1-x}\right) = \ln 16$

$\Rightarrow \frac{1+x}{1-x} = 16$

$\Rightarrow 1 + x = 16 - 16x$

$\Rightarrow 17x = 15$

$\Rightarrow x = \frac{15}{17}$

Alternative Method:

$\operatorname{artanh} x = \ln 4$

$\Rightarrow x = \tanh(\ln 4)$

$= \frac{e^{2\ln 4}-1}{e^{2\ln 4}+1}$

$= \frac{e^{\ln 16}-1}{e^{\ln 16}+1}$

$= \frac{16-1}{16+1} = \frac{15}{17}$

14) d)

$\operatorname{arsinh} 2x = 2\ln 3$

$\Rightarrow \operatorname{arsinh} 2x = \ln 9$

$\Rightarrow \ln\left(2x + \sqrt{(2x)^2 + 1}\right) = \ln 9$

$\Rightarrow 2x + \sqrt{4x^2 + 1} = 9$

$\Rightarrow \sqrt{4x^2 + 1} = 9 - 2x$

$\Rightarrow 4x^2 + 1 = (9 - 2x)^2$

$\Rightarrow 4x^2 + 1 = 4x^2 - 36x + 81$

$\Rightarrow 1 = -36x + 81$

$\Rightarrow 36x = 80$

$\Rightarrow x = \frac{20}{9}$

Alternative Method:

$\operatorname{arsinh} 2x = 2\ln 3$

$\Rightarrow \operatorname{arsinh} 2x = \ln 9$

$\Rightarrow 2x = \sinh(\ln 9)$

$\Rightarrow x = \frac{1}{2} \times \frac{e^{\ln 9} - e^{-\ln 9}}{2}$

$= \frac{e^{\ln 9} - e^{\ln \frac{1}{9}}}{4}$

$= \frac{9 - \frac{1}{9}}{4} = \frac{20}{9}$

14) e)

$\operatorname{arcosh} 3x = 3\ln 2$

$\Rightarrow \operatorname{arcosh} 3x = \ln 8$

$\Rightarrow \ln\left(3x + \sqrt{(3x)^2 - 1}\right) = \ln 8$

$\Rightarrow 3x + \sqrt{9x^2 - 1} = 8$

$\Rightarrow \sqrt{9x^2 - 1} = 8 - 3x$

$\Rightarrow 9x^2 - 1 = (8 - 3x)^2$

$\Rightarrow 9x^2 - 1 = 64 - 48x + 9x^2$

$\Rightarrow -1 = 64 - 48x$

$\Rightarrow 48x = 65$

$\Rightarrow x = \frac{65}{48}$

Alternative Method:

$\operatorname{arcosh} 3x = 3\ln 2$

$\Rightarrow \operatorname{arcosh} 3x = \ln 8$

$\Rightarrow 3x = \cosh(\ln 8)$

$\Rightarrow x = \frac{1}{3} \times \frac{e^{\ln 8} + e^{-\ln 8}}{2}$

$= \frac{e^{\ln 8} + e^{\ln \frac{1}{8}}}{6}$

$= \frac{8 + \frac{1}{8}}{6} = \frac{65}{48}$

14) f)

$\operatorname{artanh}\frac{1}{2}x = 2\ln\frac{1}{2}$

$\Rightarrow \operatorname{artanh}\frac{1}{2}x = \ln\frac{1}{4}$

$\Rightarrow \frac{1}{2}\ln\left(\frac{1+\frac{1}{2}x}{1-\frac{1}{2}x}\right) = \ln\frac{1}{4}$

$\Rightarrow \ln\left(\frac{1+\frac{1}{2}x}{1-\frac{1}{2}x}\right) = 2\ln\frac{1}{4}$

$\Rightarrow \ln\left(\frac{1+\frac{1}{2}x}{1-\frac{1}{2}x}\right) = \ln\frac{1}{16}$

$\Rightarrow \frac{1+\frac{1}{2}x}{1-\frac{1}{2}x} = \frac{1}{16}$

$\Rightarrow 16\left(1+\frac{1}{2}x\right) = 1-\frac{1}{2}x$

$\Rightarrow 16 + 8x = 1 - \frac{1}{2}x$

$\Rightarrow \frac{17}{2}x = -15$

$\Rightarrow x = -\frac{30}{17}$

Alternative Method:

$\operatorname{artanh}\frac{1}{2}x = 2\ln\frac{1}{2}$

$\Rightarrow \operatorname{artanh}\frac{1}{2}x = \ln\frac{1}{4}$

$\Rightarrow \frac{1}{2}x = \tanh\left(\ln\frac{1}{4}\right)$

$\Rightarrow x = 2\times\frac{e^{2\ln\frac{1}{4}}-1}{e^{2\ln\frac{1}{4}}+1}$

$= 2\times\frac{e^{\ln\frac{1}{16}}-1}{e^{\ln\frac{1}{16}}+1}$

$= 2\times\frac{\frac{1}{16}-1}{\frac{1}{16}+1}$

$= 2\times-\frac{15}{17} = -\frac{30}{17}$

15) a)

$x = \sinh(\ln(3-x))$

$\Rightarrow \operatorname{arsinh}(x) = \ln(3-x)$

$\Rightarrow \ln\left(x+\sqrt{x^2+1}\right) = \ln(3-x)$

$\Rightarrow x+\sqrt{x^2+1} = 3-x$

$\Rightarrow \sqrt{x^2+1} = 3-2x$

$\Rightarrow x^2+1 = (3-2x)^2$

$\Rightarrow x^2+1 = 9-12x+4x^2$

$\Rightarrow 0 = 3x^2-12x+8$

$\Rightarrow 0 = 3(x^2-4x)+8$

$\Rightarrow 0 = 3((x-2)^2-4)+8$

$\Rightarrow 0 = 3(x-2)^2-12+8$

$\Rightarrow 3(x-2)^2 = 4$

$\Rightarrow (x-2)^2 = \frac{4}{3}$

$\Rightarrow x-2 = \pm\frac{2\sqrt{3}}{3}$

$\Rightarrow x = 2\pm\frac{2\sqrt{3}}{3}$

$x = 2-\frac{2\sqrt{3}}{3}$ only as $\ln(3-x)$ is undefined when $x = 2+\frac{2\sqrt{3}}{3}$

Alternative Method:

$x = \sinh(\ln(3-x))$

$\Rightarrow x = \frac{e^{\ln(3-x)}-e^{-\ln(3-x)}}{2}$

$\Rightarrow 2x = e^{\ln(3-x)}-e^{\ln(3-x)^{-1}}$

$\Rightarrow 2x = 3-x-\frac{1}{3-x}$

$\Rightarrow \frac{1}{3-x} = 3-3x$

$\Rightarrow 1 = (3-x)(3-3x)$

$\Rightarrow 1 = 9-12x+3x^2$

$\Rightarrow 0 = 3x^2-12x+8$

$\Rightarrow 0 = 3(x^2-4x)+8$

$\Rightarrow 0 = 3((x-2)^2-4)+8$

$\Rightarrow 0 = 3(x-2)^2-12+8$

$\Rightarrow 3(x-2)^2 = 4$

$\Rightarrow (x-2)^2 = \frac{4}{3}$

$\Rightarrow x-2 = \pm\frac{2\sqrt{3}}{3}$

$\Rightarrow x = 2 \pm \frac{2\sqrt{3}}{3}$

$x = 2 - \frac{2\sqrt{3}}{3}$ only as $\ln(3 - x)$ is undefined when $x = 2 + \frac{2\sqrt{3}}{3}$

15) b)

$\cosh(\ln(2x - 1)) = x + 2$

$\Rightarrow \ln(2x - 1) = \operatorname{arcosh}(x + 2)$

$\Rightarrow \ln(2x - 1) = \ln\left(x + 2 + \sqrt{(x + 2)^2 - 1}\right)$

$\Rightarrow 2x - 1 = x + 2 + \sqrt{x^2 + 4x + 3}$

$\Rightarrow x - 3 = \sqrt{x^2 + 4x + 3}$

$\Rightarrow (x - 3)^2 = x^2 + 4x + 3$

$\Rightarrow x^2 - 6x + 9 = x^2 + 4x + 3$

$\Rightarrow -6x + 9 = 4x + 3$

$\Rightarrow 6 = 10x$

$\Rightarrow x = \frac{3}{5}$

Alternative Method:

$\cosh(\ln(2x - 1)) = x + 2$

$\Rightarrow \frac{e^{\ln(2x-1)} + e^{-\ln(2x-1)}}{2} = x + 2$

$\Rightarrow e^{\ln(2x-1)} + e^{\ln(2x-1)^{-1}} = 2x + 4$

$\Rightarrow 2x - 1 + \frac{1}{2x-1} = 2x + 4$

$\Rightarrow \frac{1}{2x-1} = 5$

$\Rightarrow 2x - 1 = \frac{1}{5}$

$\Rightarrow x = \frac{3}{5}$

15) c)

$x-1=\tanh\left(\ln\sqrt{x+1}\right)$

$\Rightarrow \operatorname{artanh}(x-1)=\ln\left(\sqrt{x+1}\right)$

$\Rightarrow \frac{1}{2}\ln\left(\frac{1+(x-1)}{1-(x-1)}\right)=\frac{1}{2}\ln(x+1)$

$\Rightarrow \ln\left(\frac{x}{2-x}\right)=\ln(x+1)$

$\Rightarrow \frac{x}{2-x}=x+1$

$\Rightarrow x=(x+1)(2-x)$

$\Rightarrow x=-x^2+x+2$

$\Rightarrow x^2=2$

$x=\sqrt{2}$ only as $\sqrt{x+1}$ is not real when $x=-\sqrt{2}$

Alternative Method:

$x-1=\tanh\left(\ln\sqrt{x+1}\right)$

$\Rightarrow x-1=\frac{e^{2\ln\sqrt{x+1}}-1}{e^{2\ln\sqrt{x+1}}+1}$

$\Rightarrow x-1=\frac{e^{\ln(x+1)}-1}{e^{\ln(x+1)}+1}$

$\Rightarrow x-1=\frac{x+1-1}{x+1+1}$

$\Rightarrow x-1=\frac{x}{x+2}$

$\Rightarrow (x-1)(x+2)=x$

$\Rightarrow x^2+x-2=x$

$\Rightarrow x^2=2$

$x=\sqrt{2}$ only as $\sqrt{x+1}$ is not real when $x=-\sqrt{2}$

16) a)

$\operatorname{artanh} x - \ln(1+x) = \ln\frac{6}{5}$

$\frac{1}{2}\ln\left(\frac{1+x}{1-x}\right) - \ln(1+x) = \ln\frac{6}{5}$

$\Rightarrow \ln\left(\frac{1+x}{1-x}\right) - 2\ln(1+x) = 2\ln\frac{6}{5}$

$\Rightarrow \ln\left(\frac{1+x}{1-x}\right) - \ln(1-x)^2 = \ln\frac{36}{25}$

$\Rightarrow \ln\left(\frac{1+x}{1-x} \times \frac{1}{(1+x)^2}\right) = \ln\frac{36}{25}$

$\Rightarrow \frac{1}{(1-x)(1+x)} = \frac{36}{25}$

$\Rightarrow (1-x)(1+x) = \frac{25}{36}$

$\Rightarrow 1 - x^2 = \frac{25}{36}$

$\Rightarrow x^2 = \frac{11}{36}$

$\Rightarrow x = \pm\frac{\sqrt{11}}{6}$

16) b)

$\ln 2 + \ln(1+x) - \operatorname{artanh} x = \ln\frac{2}{9}$

$\Rightarrow \ln\big(2(1+x)\big) - \frac{1}{2}\ln\left(\frac{1+x}{1-x}\right) = \ln\frac{2}{9}$

$\Rightarrow 2\ln\big(2(1+x)\big) - \ln\left(\frac{1+x}{1-x}\right) = 2\ln\frac{2}{9}$

$\Rightarrow \ln(4(1+x)^2) - \ln\left(\frac{1+x}{1-x}\right) = \ln\frac{4}{81}$

$\Rightarrow \ln\left(4(1+x)^2 \times \frac{1-x}{1+x}\right) = \ln\frac{4}{81}$

$\Rightarrow 4(1+x)(1-x) = \frac{4}{81}$

$\Rightarrow 1 - x^2 = \frac{1}{81}$

$\Rightarrow x^2 = \frac{80}{81}$

$\Rightarrow x = \pm\frac{4\sqrt{5}}{9}$

16) c)

$\operatorname{arsinh} x - \ln(1+x) = \ln 2$

$\Rightarrow \ln\left(x+\sqrt{x^2+1}\right) - \ln(1+x) = \ln 2$

$\Rightarrow \ln\left(x+\sqrt{x^2+1}\right) = \ln 2 + \ln(1+x)$

$\Rightarrow \ln\left(x+\sqrt{x^2+1}\right) = \ln(2+2x)$

$\Rightarrow x+\sqrt{x^2+1} = 2+2x$

$\Rightarrow \sqrt{x^2+1} = 2+x$

$\Rightarrow x^2+1 = (2+x)^2$

$\Rightarrow x^2+1 = 4+4x+x^2$

$\Rightarrow 1 = 4+4x$

$\Rightarrow 4x = -3$

$\Rightarrow x = -\frac{3}{4}$

16) d)

$\ln 3 + \ln(x-1) - \operatorname{arcosh} x = 0$

$\Rightarrow \ln(3x-3) - \ln\left(x+\sqrt{x^2-1}\right) = 0$

$\Rightarrow \ln(3x-3) = \ln\left(x+\sqrt{x^2-1}\right)$

$\Rightarrow 3x-3 = x+\sqrt{x^2-1}$

$\Rightarrow 2x-3 = \sqrt{x^2-1}$

$\Rightarrow (2x-3)^2 = x^2-1$

$\Rightarrow 4x^2-12x+9 = x^2-1$

$\Rightarrow 3x^2-12x+10 = 0$

$\Rightarrow 3(x^2-4x)+10 = 0$

$\Rightarrow 3((x-2)^2-4)+10 = 0$

$\Rightarrow 3(x-2)^2-12+10 = 0$

$\Rightarrow (x-2)^2 = \frac{2}{3}$

$\Rightarrow x-2 = \pm\sqrt{\frac{2}{3}}$

$\Rightarrow x = 2 \pm \sqrt{\frac{2}{3}}$

$x = 2+\sqrt{\frac{2}{3}}$ only as $x = 2-\sqrt{\frac{2}{3}}$ does not solve the original equation.

17) a)

$\operatorname{arsinh} x + \ln 2 = \operatorname{arcosh} x + \ln 3$

$\Rightarrow \ln\left(x + \sqrt{x^2+1}\right) + \ln 2 = \ln\left(x + \sqrt{x^2-1}\right) + \ln 3$

$\Rightarrow \ln\left(2x + 2\sqrt{x^2+1}\right) = \ln\left(3x + 3\sqrt{x^2-1}\right)$

$\Rightarrow 2x + 2\sqrt{x^2+1} = 3x + 3\sqrt{x^2-1}$

$\Rightarrow 2\sqrt{x^2+1} - 3\sqrt{x^2-1} = x$

$\Rightarrow \left(2\sqrt{x^2+1} - 3\sqrt{x^2-1}\right)^2 = x^2$

$\Rightarrow 4(x^2+1) - 12\sqrt{x^4-1} + 9(x^2-1) = x^2$

$\Rightarrow 4x^2 + 4 + 9x^2 - 9 - x^2 = 12\sqrt{x^4-1}$

$\Rightarrow 12x^2 - 5 = 12\sqrt{x^4-1}$

$\Rightarrow (12x^2-5)^2 = 144(x^4-1)$

$\Rightarrow 144x^4 - 120x^2 + 25 = 144x^4 - 144$

$\Rightarrow -120x^2 + 25 = -144$

$\Rightarrow 120x^2 = 169$

$\Rightarrow x^2 = \frac{169}{120}$

$\Rightarrow x = \pm\frac{13\sqrt{30}}{60}$

$x = \frac{13\sqrt{30}}{60}$ only as $x \geq 1$ for $\operatorname{arcosh} x$

17) b)

$\operatorname{arsinh} x - 2\operatorname{arcosh} x = 0$

$\Rightarrow \ln\left(x + \sqrt{x^2+1}\right) - 2\ln\left(x + \sqrt{x^2-1}\right) = 0$

$\Rightarrow \ln\left(x + \sqrt{x^2+1}\right) = \ln\left(x + \sqrt{x^2-1}\right)^2$

$\Rightarrow x + \sqrt{x^2+1} = \left(x + \sqrt{x^2-1}\right)^2$

$\Rightarrow x + \sqrt{x^2+1} = x^2 + 2x\sqrt{x^2-1} + x^2 - 1$

$\Rightarrow \sqrt{x^2+1} - 2x\sqrt{x^2-1} = 2x^2 - x - 1$

$\Rightarrow \left(\sqrt{x^2+1} - 2x\sqrt{x^2-1}\right)^2 = (2x^2 - x - 1)^2$

$\Rightarrow x^2 + 1 - 4x\sqrt{x^4-1} + 4x^2(x^2-1) = 4x^4 - 4x^3 - 3x^2 + 2x + 1$

$\Rightarrow 4x^3 - 2x = 4x\sqrt{x^4-1}$

$\Rightarrow 2x^2 - 1 = 2\sqrt{x^4-1}$ as $x \neq 0$

$\Rightarrow (2x^2-1)^2 = 4(x^4-1)$

$\Rightarrow 4x^4 - 4x^2 + 1 = 4x^4 - 4$

$\Rightarrow -4x^2 + 1 = -4$

$\Rightarrow 4x^2 = 5$

$\Rightarrow x^2 = \frac{5}{4}$

$\Rightarrow x = \pm\frac{\sqrt{5}}{2}$

$x = \frac{\sqrt{5}}{2}$ only as $x \geq 1$ for $\operatorname{arcosh} x$